

Q-List two advantages and two disadvantages of static routing compared to dynamic routing, using examples from real-life apps like IRCTC or Swiggy to illustrate your points.

Hint: Think about how static routes might impact delivery or booking updates in these apps.

Point	Static Routing	Dynamic Routing
Advantage 1: Simple and predictable	Routes are manually configured, so the network administrator knows exactly where traffic will go. For example, a small IRCTC branch network could use fixed routes between its booking server and office network.	Dynamic routing automatically learns and updates routes.
Advantage 2: Low overhead	Static routes do not exchange routing information continuously, so they use less CPU, memory, and bandwidth. This can be useful for a small Swiggy delivery-office network where the network paths rarely change.	Dynamic routing requires routing protocols to exchange information.
Disadvantage 1: Manual updates	If a network link fails or its address changes, an administrator must manually change the route. For example, if an IRCTC server connection changes and the static route is not updated, ticket-booking requests or status updates could fail or be delayed.	Dynamic routing can automatically find an alternative path.
Disadvantage 2: Poor scalability	Managing hundreds of branches manually becomes difficult. For example, if Swiggy has many city/restaurant networks, maintaining static routes for every location could make delivery-status updates and order communication harder to manage when routes change.	Dynamic routing is more suitable for large, changing networks.